5: Implementation bagging using Random Forests

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# Import libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# Load Iris dataset
iris = load_iris()
X = pd.DataFrame(iris.data, columns=iris.feature_names)
y = pd.Series(iris.target)

# Show first 5 rows
print("Sample of Iris Dataset:")
print(X.head())

# Visualize data distribution
sns.pairplot(pd.concat([X, pd.Series(y, name="Species")], axis=1),
hue="Species", palette="Set2")
plt.suptitle(" Iris Feature Pairplot", y=1.02)
plt.show()

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)

# Create Random Forest (Bagging)
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy and report
accuracy = accuracy_score(y_test, y_pred)
print(f"\n Model Accuracy: {accuracy:.2f}")
print("\n Classification Report:\n", classification_report(y_test, y_pred,
target_names=iris.target_names))

# Confusion matrix heatmap
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, cmap="Blues", fmt="d",
```

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             xticklabels=iris.target_names,
             yticklabels=iris.target_names)
plt.title(" Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Feature importance visualization
feature_importance = pd.Series(model.feature_importances_,
index=X.columns)
feature_importance.sort_values().plot(kind='barh', color='skyblue')
plt.title(" Feature Importance from Random Forest")
plt.xlabel("Importance")
plt.ylabel("Feature")
plt.show()
```
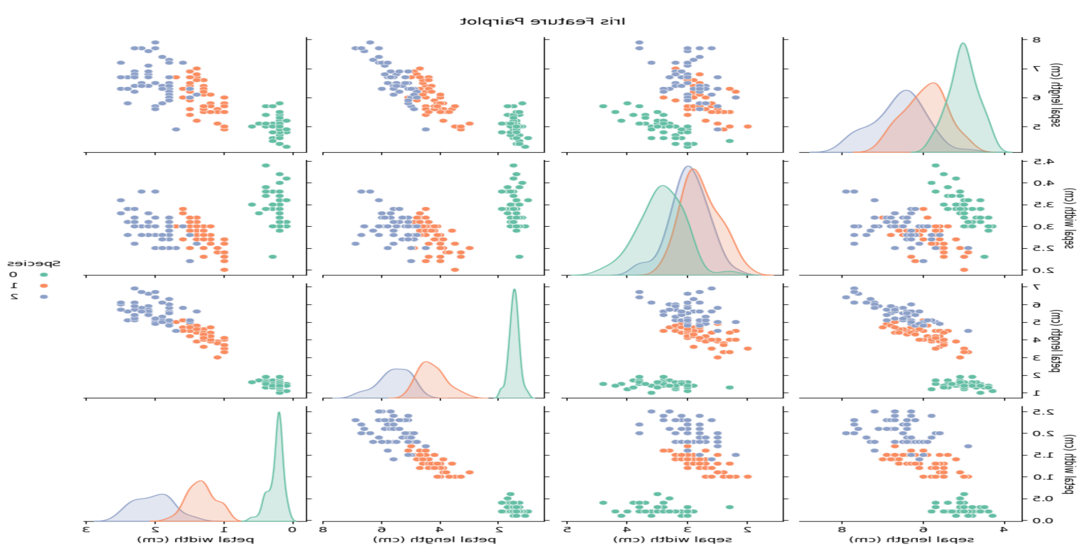

Output:

```
Sample of Iris Dataset:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2
```




```
Model Accuracy: 1.00

Classification Report:
              precision    recall  f1-score   support
```

setosa	1.00	1.00	1.00	19
versicolor	1.00	1.00	1.00	13
virginica	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45